Ahmed Magdy

AI/ML Engineer

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LinkedIn | GitHub | Kaggle | Huggingface

Summary

Computer Science graduate specializing in AI/ML engineering, LLM orchestration, and backend architecture. Experienced in building asynchronous processing pipelines, agentic workflows, RAG systems, and delivering end-to-end model deployments. Strong foundation in software engineering, OOP, Data Structures, Algorithms and distributed systems.

TECHNICAL SKILLS

- **AI & Generative AI: Large Language Models (LLMs), RAG Architectures, Vector Databases (ChromaDB), Agentic Workflows (LangChain/LangGraph), Prompt Engineering, Fine-Tuning (PEFT/LoRA concepts), Audio Signal Processing (ASR/SER), NLP, Computer Vision.**
- **Frameworks & Libraries: PyTorch, TensorFlow, Keras, Hugging Face, LLMs, Scikit-Learn, Librosa, SoundFile, NumPy, Pandas, Matplotlib.**
- **Backend & DevOps: FastAPI, Uvicorn, Docker, Docker Compose, Asynchronous Pipelines, Ngrok.**
- **Programming Languages: Python, C++, C, Java (OOP Foundation).**
- **Databases & Tools: SQL, SQL Server, Git, GitHub, Microsoft Office Suite.**

PROJECTS

AI/ML Multimodal Audio Engine & Distributed Backend Pipeline (Graduation Project)

- **Architecture Overview: Designed and deployed a multimodal psychiatric analytics framework (PACE) that processes raw Egyptian-Arabic psychotherapy session recordings to track patient emotional timelines and automate structured clinical diagnostic reports.**
- **Asynchronous Model Serving: Developed a high-performance RESTful API using FastAPI and Uvicorn to serve large-scale deep learning models as decoupled microservices.**
- **Concurrency & Task Queueing: Integrated asynchronous task queues via FastAPI BackgroundTasks to manage heavy multi-model inference pipelines, eliminating server-side timeouts and preserving event-loop responsiveness.**
- **Memory Optimization & Chunking: Designed an in-memory audio-slicing pipeline using Librosa and SoundFile to dynamically stream continuous 30-second windows, drastically reducing peak VRAM footprint and preventing Out-Of-Memory (OOM) crashes.**
- **Multi-GPU Late-Fusion & RAG Pipeline: Orchestrated a multi-GPU pipeline distributing workloads across hardware—integrating offline Egyptian-Arabic Whisper, fine-tuned Wav2Vec2, and CAMeLBERT, routed into a localized Qwen LLM with Context Retrieval (RAG) for clinical reports.**

Retrieval-Augmented Generation (RAG) & Agentic Knowledge Assistant

- **Data Ingestion & Vector Indexing: Built an automated ETL pipeline to parse, chunk, and embed enterprise documents into a Vector Database (ChromaDB) for high-precision semantic search.**
- **Agentic Orchestration: Developed an interactive AI Agent utilizing LangChain / LangGraph for multi-step reasoning, tool calling, and dynamic query routing to query structured SQL databases and unstructured vector stores.**
- **API Deployment: Exposed the agentic RAG pipeline as a production-ready RESTful service using FastAPI, implementing strict latency and groundedness evaluation checks.**

(HelloML) Optical Character Recognition (OCR)

- **OCR Pipeline Engineering: Built an Optical Character Recognition (OCR) application from scratch to classify handwritten digits (0-9) using foundational Machine Learning algorithms.**
- **Image Processing & Features: Designed the end-to-end pipeline including digital image preprocessing, noise reduction, and pixel-grid feature extraction to ensure high classification accuracy.**
- **Low-Level Data Parsing: Refactored a separate heat diffusion simulation matrix, replacing random generation with logic that parses and ingests raw structured matrix data directly from local binary storage files.**

Web Scraping & Algorithmic Search Engine

- **Data Ingestion & Scraping: Engineered a robust web scraping pipeline using BeautifulSoup/Scrapy to extract, clean, and preprocess large-scale unstructured textual data from targeted web sources.**
- **Search Architecture: Developed a custom search and retrieval engine incorporating text indexing, algorithmic filtering, and ranking mechanisms to query processed data efficiently.**
- **Backend Integration: Integrated the data pipeline with a Python backend (Flask/Django), ensuring optimized query execution and seamless data flow from storage to the end-user.**

EDUCATION

Fayoum University | Faculty of Computers and Artificial Intelligence 2022-2026

LANGUAGES

Arabic: Native speaker.

English: Excellent / Professional proficiency.

Courses & Certificates

- **Coursera: NLP & Machine Learning & Deep Learning Specializations (Andrew Ng).**
- **ITI: Applied Deep Learning**
- **ITI: Mastering Object-Oriented Programming (OOP) using C++.**
- **ITI: Computer Network Fundamentals | C Programming Mastery**